

101. Which of the following compounds volatilises on heating

- (a) MgCl_2
- (b) HgCl_2
- (c) CaCl_2
- (d) FeCl_3

Answer: B — (b) HgCl_2

Chemical reason: Applying the known reaction of the stated transition-metal compound under these reagents/conditions gives HgCl_2 . The choice is consistent with the expected oxidation state and stability of the product.

102. Which of the following statement is not true

- (a) Colourless compounds of transition elements are paramagnetic
- (b) Coloured compounds of transition elements are paramagnetic
- (c) Colourless compounds of transition elements are diamagnetic
- (d) Transition elements form the complex compounds

Answer: A — (a) Colourless compounds of transition elements are paramagnetic

Chemical reason: The correct choice is Colourless compounds of transition elements are paramagnetic. It matches the established composition, oxidation state, and characteristic chemical behaviour of the substance named in the question; the other options are inconsistent with one or more of these features.

103. Calamine is a mineral, which is

- (a) ZnCO_3
- (b) ZnS
- (c) ZnSO_4
- (d) ZnO

Answer: A — (a) ZnCO_3

Chemical reason: Calamine is a zinc mineral traditionally associated with ZnCO_3 (smithsonite) and related zinc silicate ores; the correct option is the zinc ore formula/name.

104. Super conductors are derived from compounds of

- (a) p-block elements
- (b) Lanthanides
- (c) Actinides
- (d) Transition elements

Answer: D — (d) Transition elements

Chemical reason: The correct choice is Transition elements. It matches the established composition, oxidation state, and characteristic chemical behaviour of the substance named in the question; the other options are inconsistent with one or more of these features.

105. Manganese achieves its highest oxidation state in its compound

- (a) MnO_3
- (b) Mn_3O_4
- (c) KMnO_4
- (d) K_2MnO_4

Answer: C — (c) KMnO_4

Chemical reason: The result is obtained by assigning oxidation numbers before and after the



reaction and balancing electron transfer. The species in KMnO_4 has the oxidation-state change required by the stated redox conditions, so that choice is correct.

106. Which can be reduced to the metal by heating it in a stream of hydrogen

- (a) Copper (II) oxide
- (b) Magnesium oxide
- (c) Aluminium oxide
- (d) Calcium oxide

Answer: A — (a) Copper (II) oxide

Chemical reason: The result is obtained by assigning oxidation numbers before and after the reaction and balancing electron transfer. The species in Copper (II) oxide has the oxidation-state change required by the stated redox conditions, so that choice is correct.

107. Which of the following is coloured

- (a) ScCl_3
- (b) TiO_2
- (c) MnSO_4
- (d) ZnSO_4

Answer: C — (c) MnSO_4

Chemical reason: The colour follows from the electronic structure and coordination environment of the transition-metal ion. The species in MnSO_4 has the characteristic oxidation state/ligand field that produces the observed colour.

108. Chrome green is

- (a) Chromium sulphate
- (b) Chromium chloride
- (c) Chromium nitrate
- (d) Chromium oxide

Answer: D — (d) Chromium oxide

Chemical reason: Chrome green is chromium(III) oxide, Cr_2O_3 . It is a stable green pigment because Cr^{3+} in the oxide lattice strongly absorbs complementary visible wavelengths.

109. The colour of $(\text{NH}_4)_2\text{SO}_4\text{Fe}(\text{SO}_4)_3 \cdot 24\text{H}_2\text{O}$ is

- (a) White
- (b) Green
- (c) Violet
- (d) Blue

Answer: B — (b) Green

Chemical reason: The colour follows from the electronic structure and coordination environment of the transition-metal ion. The species in Green has the characteristic oxidation state/ligand field that produces the observed colour.

110. Correct formula of potassium ferricyanide is

- (a) $\text{K}_4\text{Fe}(\text{CN})_6$
- (b) $\text{K}_3\text{Fe}(\text{CN})_6$
- (c) $\text{K}_3[\text{Fe}(\text{CN})_3]$
- (d) $\text{K}_3[\text{Fe}(\text{CN})_4]$

Answer: B — (b) $\text{K}_3\text{Fe}(\text{CN})_6$

Chemical reason: Potassium ferricyanide is $\text{K}_3[\text{Fe}(\text{CN})_6]$. The complex anion $[\text{Fe}(\text{CN})_6]^{3-}$ requires three K^+ ions for electrical neutrality.



111. The form of iron having the highest carbon content is

- (a) Cast iron
- (b) Wrought iron
- (c) Strain steel
- (d) Mild steel

Answer: A — (a) Cast iron

Chemical reason: Cast iron has the highest carbon content among the common commercial forms listed, typically around 2–4% C, much more than steel or wrought iron.

112. Aqueous solution of ferric chloride is

- (a) Acidic
- (b) Basic
- (c) Neutral
- (d) Amphoteric

Answer: A — (a) Acidic

Chemical reason: $\text{FeCl}_3(\text{aq})$ is acidic because Fe^{3+} has high charge density and hydrolyses coordinated water, releasing H^+ . Therefore its aqueous solution has pH below 7.

113. In the reduction of dichromate by Fe(II) the number of electrons involved per chromium atom is

- (a) 2
- (b) 3
- (c) 4
- (d) 1

Answer: B — (b) 3

Chemical reason: Each Cr atom in dichromate changes from +6 to +3, so each chromium gains 3

electrons. The full $\text{Cr}_2\text{O}_7^{2-}$ ion therefore gains 6 electrons in total.

114. A group of acidic oxide is

- (a) $\text{CrO}_3, \text{Mn}_2\text{O}_7$
- (b) $\text{ZnO}, \text{Al}_2\text{O}_3$
- (c) CaO, ZnO
- (d) $\text{Na}_2\text{O}, \text{Al}_2\text{O}_3$

Answer: A — (a) $\text{CrO}_3, \text{Mn}_2\text{O}_7$

Chemical reason: The correct choice is $\text{CrO}_3, \text{Mn}_2\text{O}_7$. It matches the established composition, oxidation state, and characteristic chemical behaviour of the substance named in the question; the other options are inconsistent with one or more of these features.

115. Silver nitrate is mainly used

- (a) In photography
- (b) In model formation
- (c) As reducing agent
- (d) As dehydrating agent

Answer: A — (a) In photography

Chemical reason: Silver nitrate is an important starting material for preparing photosensitive silver halides such as AgBr , so its major classical use is in photographic chemistry.



116. The correct order of magnetic moments (spin only values in B.M.) among is

- (a) $[\text{Fe}(\text{CN})_6]_4 \rightarrow [\text{MnCl}_4]_2 \rightarrow [\text{CoCl}_4]^{2-}$
- (b) $[\text{MnCl}_4]_2 \rightarrow [\text{Fe}(\text{CN})_6]_4 \rightarrow [\text{CoCl}_4]^{2-}$
- (c) $[\text{MnCl}_4]_2 \rightarrow [\text{CoCl}_4]_2 \rightarrow [\text{Fe}(\text{CN})_6]^{4-}$
- (d) $[\text{Fe}(\text{CN})_6]_4 \rightarrow [\text{CoCl}_4]_2 \rightarrow [\text{MnCl}_4]^{2-}$ (Atomic nos. Mn=25, Fe=26, Co=27)

Answer: C — (c) $[\text{MnCl}_4]^{2-} \rightarrow [\text{CoCl}_4]^{2-} \rightarrow [\text{Fe}(\text{CN})_6]^{4-}$

Chemical reason: Magnetic behaviour depends on the number of unpaired d electrons. Writing the metal-ion d configuration for the species in $[\text{MnCl}_4]^{2-} \rightarrow [\text{CoCl}_4]^{2-} \rightarrow [\text{Fe}(\text{CN})_6]^{4-}$ gives the required number of unpaired electrons, so it has the stated magnetic behaviour.

117. Hybridization of $[\text{Ni}(\text{CO})_4]$ is

- (a) sp^3
- (b) d^2sp^3
- (c) sp^3d
- (d) sp^2

Answer: A — (a) sp^3

Chemical reason: The correct choice is sp^3 . It matches the established composition, oxidation state, and characteristic chemical behaviour of the substance named in the question; the other options are inconsistent with one or more of these features.

118. What is the oxidation number of iron in the compound $[\text{Fe}(\text{H}_2\text{O})_5(\text{NO})]\text{SO}_4$

- (a) +2
- (b) +3
- (c) ± 1
- (d) ± 4

Answer: A — (a) +2

Chemical reason: The result is obtained by assigning oxidation numbers before and after the reaction and balancing electron transfer. The species in +2 has the oxidation-state change required by the stated redox conditions, so that choice is correct.

119. Which of the following metal gives hydrogen gas, when heated with hot concentrated alkali

- (a) Cu
- (b) Ag
- (c) Zn
- (d) Ni

Answer: C — (c) Zn

Chemical reason: Zinc is amphoteric and reacts with hot concentrated alkali to form zincate while liberating H_2 . Cu, Ag, and Ni do not show this characteristic reaction under the same conditions.





120. When ferric oxide reacts with NaOH, the product formed is

- (a) NaF
- (b) FeCl_3
- (c) Fe(OH)_3
- (d) NaFeO_2

Answer: C — (c) Fe(OH)_3

Chemical reason: Amphoteric metals/oxides react with both acids and strong bases, often forming soluble hydroxo or oxo complexes in alkali. The species/result Fe(OH)_3 is consistent with that behaviour.

